

Report on

ETL Process with SSIS

Prepared By:

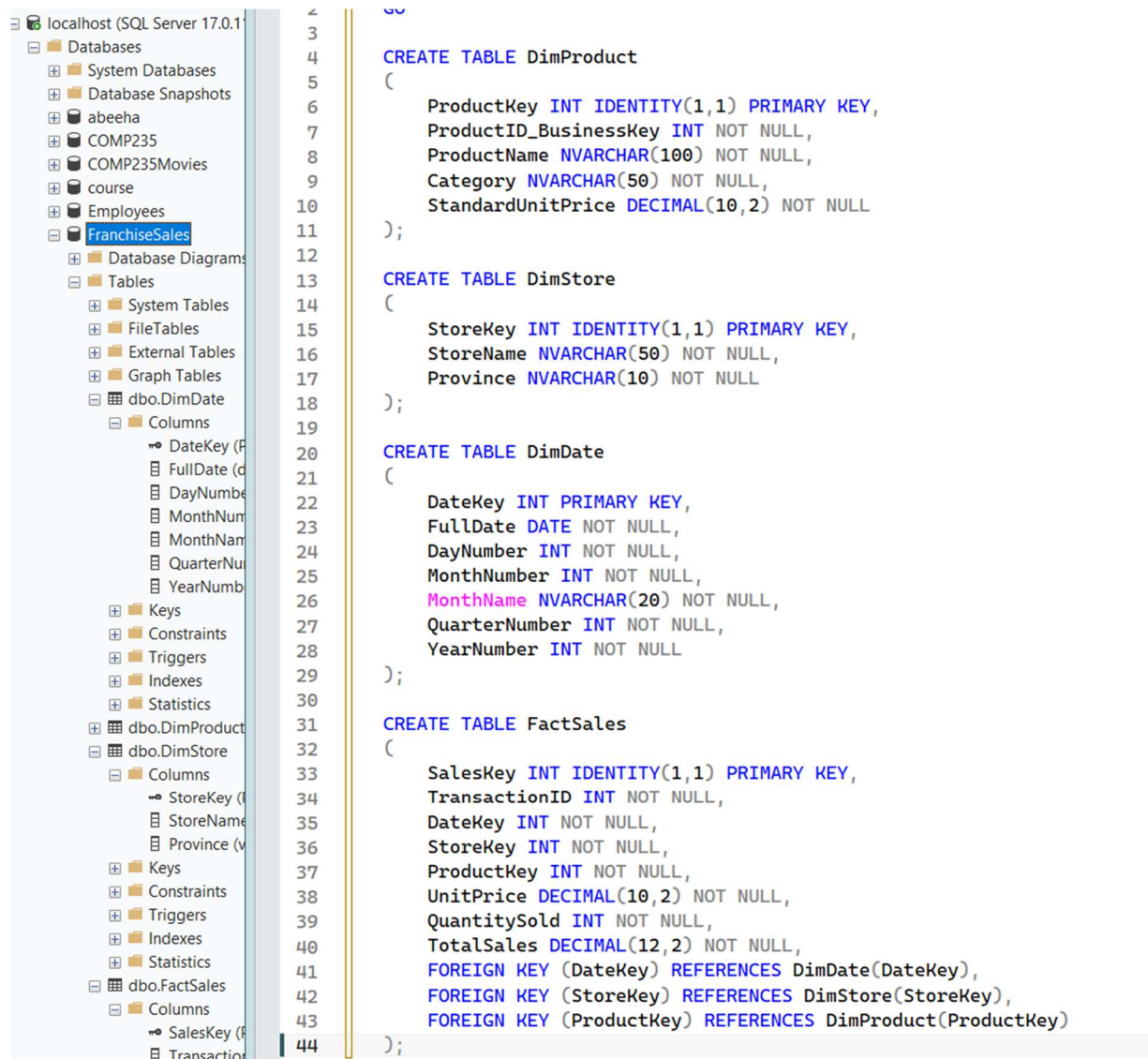
01. Md Alamgir Hossain

02. Md Moksudul Haque

Part 1: SSMS Database Setup

Database and 4 table creation

Created the data warehouse database and defined all dimension and fact tables required for the ETL process.



The screenshot displays the SQL Server Enterprise Manager (SSMS) interface. On the left, the 'Databases' folder is expanded, showing the 'FranchiseSales' database. Under 'Tables', the following tables are listed: 'dbo.DimDate', 'dbo.DimProduct', 'dbo.DimStore', and 'dbo.FactSales'. The 'Columns' and 'Keys' sections are also visible for each table. On the right, the SQL script for creating these tables is shown. The script includes the following statements:

```
CREATE TABLE DimProduct
(
    ProductKey INT IDENTITY(1,1) PRIMARY KEY,
    ProductID_BusinessKey INT NOT NULL,
    ProductName NVARCHAR(100) NOT NULL,
    Category NVARCHAR(50) NOT NULL,
    StandardUnitPrice DECIMAL(10,2) NOT NULL
);

CREATE TABLE DimStore
(
    StoreKey INT IDENTITY(1,1) PRIMARY KEY,
    StoreName NVARCHAR(50) NOT NULL,
    Province NVARCHAR(10) NOT NULL
);

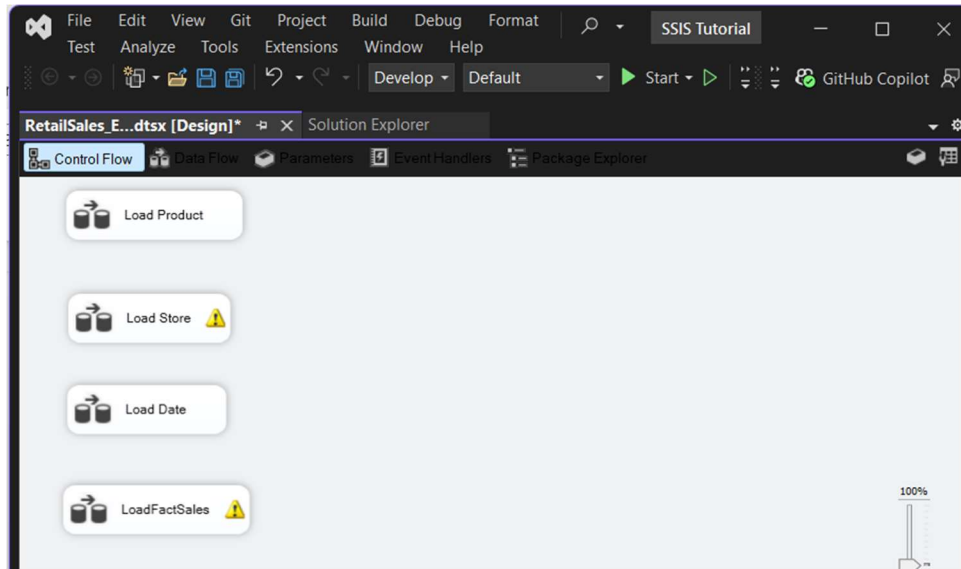
CREATE TABLE DimDate
(
    DateKey INT PRIMARY KEY,
    FullDate DATE NOT NULL,
    DayNumber INT NOT NULL,
    MonthNumber INT NOT NULL,
    MonthName NVARCHAR(20) NOT NULL,
    QuarterNumber INT NOT NULL,
    YearNumber INT NOT NULL
);

CREATE TABLE FactSales
(
    SalesKey INT IDENTITY(1,1) PRIMARY KEY,
    TransactionID INT NOT NULL,
    DateKey INT NOT NULL,
    StoreKey INT NOT NULL,
    ProductKey INT NOT NULL,
    UnitPrice DECIMAL(10,2) NOT NULL,
    QuantitySold INT NOT NULL,
    TotalSales DECIMAL(12,2) NOT NULL,
    FOREIGN KEY (DateKey) REFERENCES DimDate(DateKey),
    FOREIGN KEY (StoreKey) REFERENCES DimStore(StoreKey),
    FOREIGN KEY (ProductKey) REFERENCES DimProduct(ProductKey)
);
```

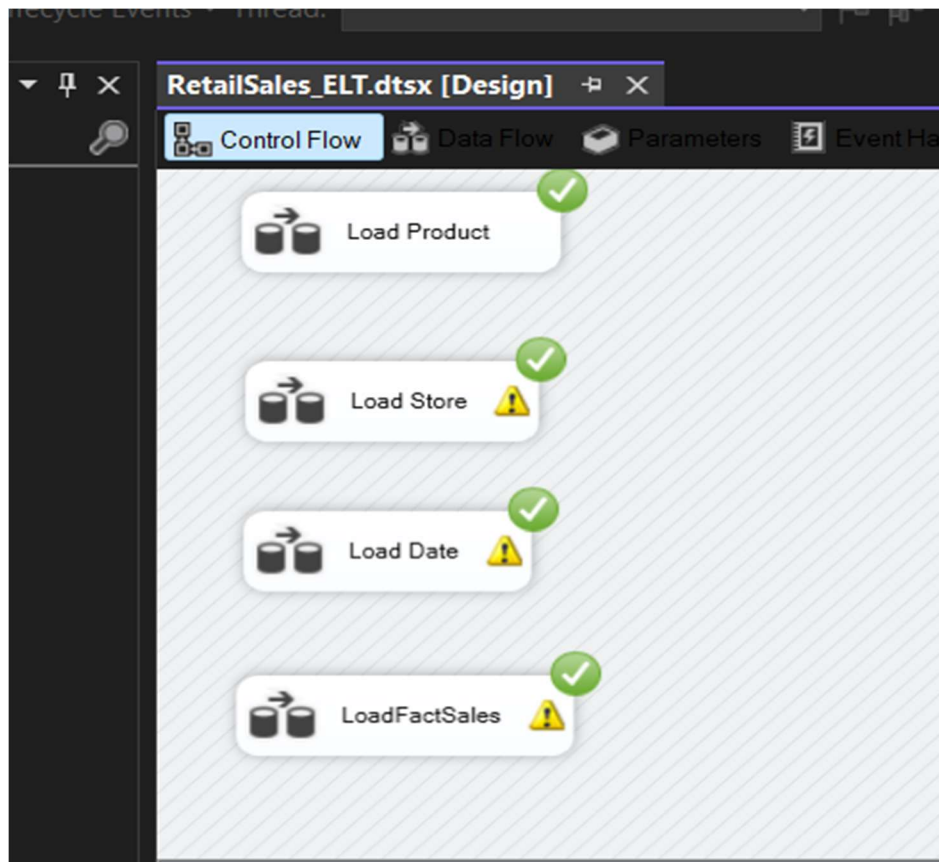
Part 2: Visual Studio Project Setup

2.1 Load Product

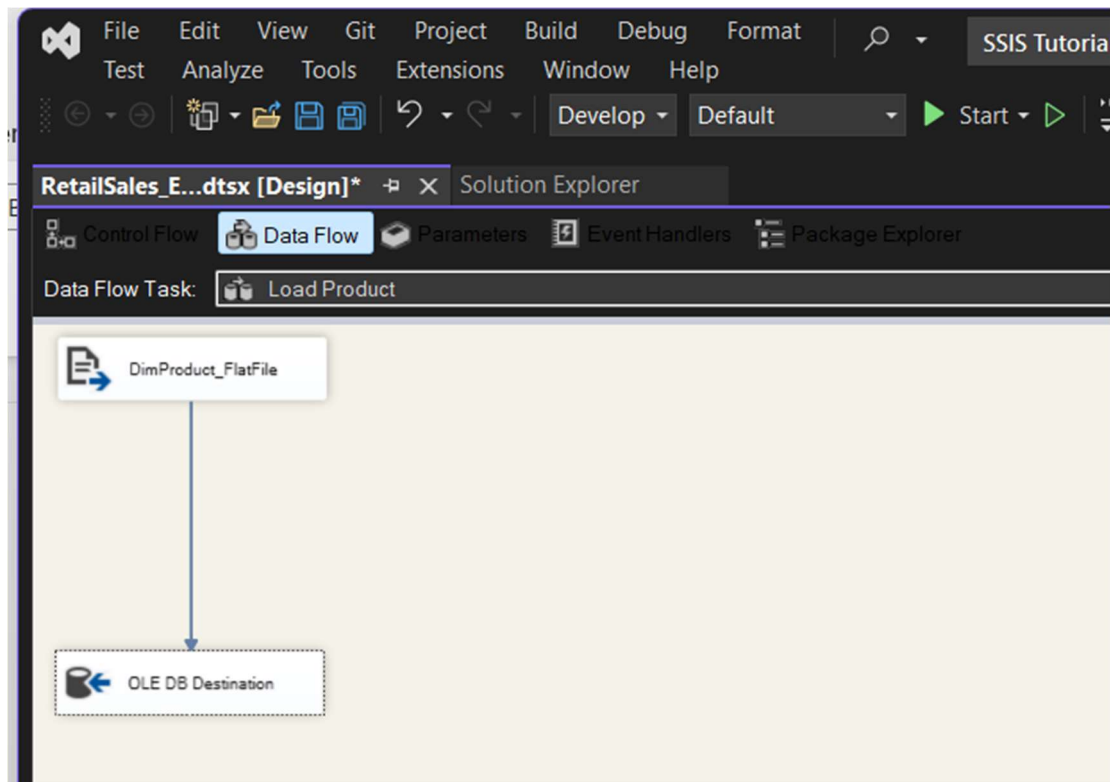
Configured a data flow task to load product data from a flat file into the DimProduct table.



2.2 All Control Flow run successfully.

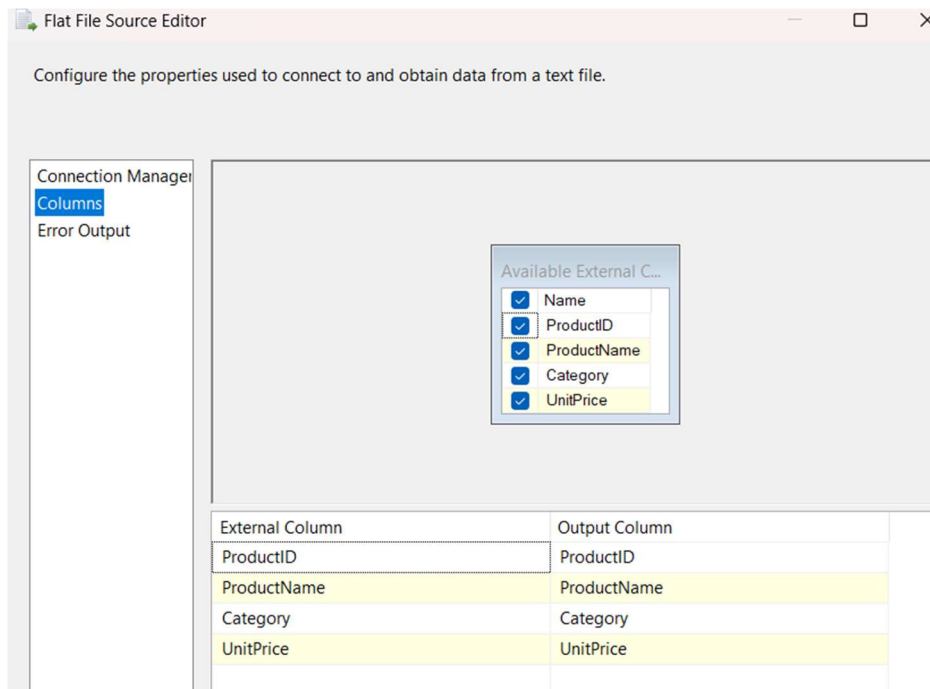


2.1.1 Control Flow-> Load Product:



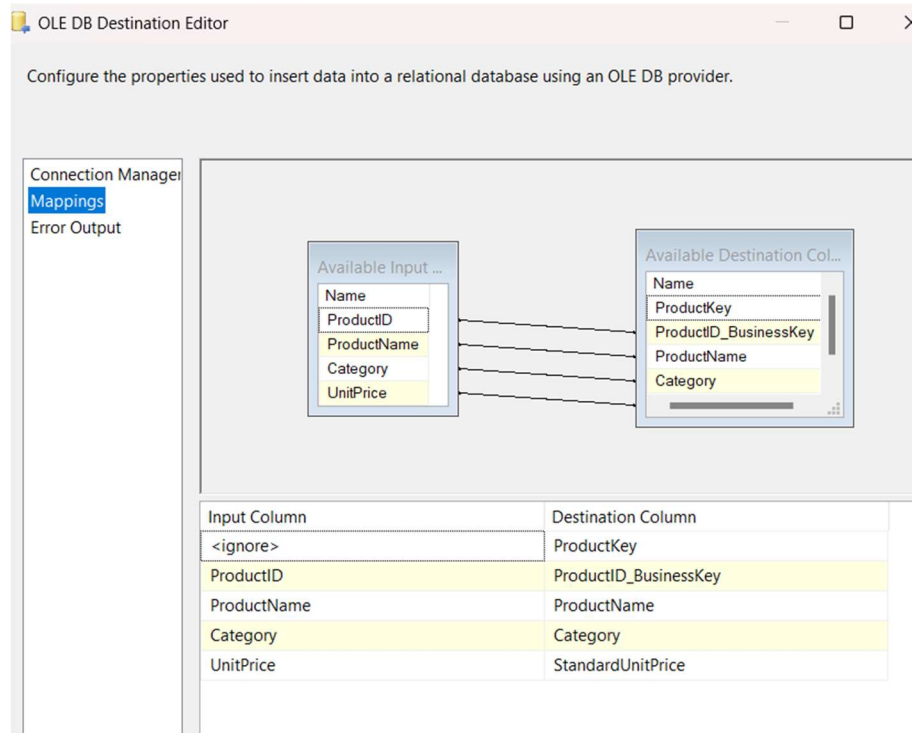
2.2 Column Display

Verified and reviewed all columns from the source file to ensure correct data types and structure.



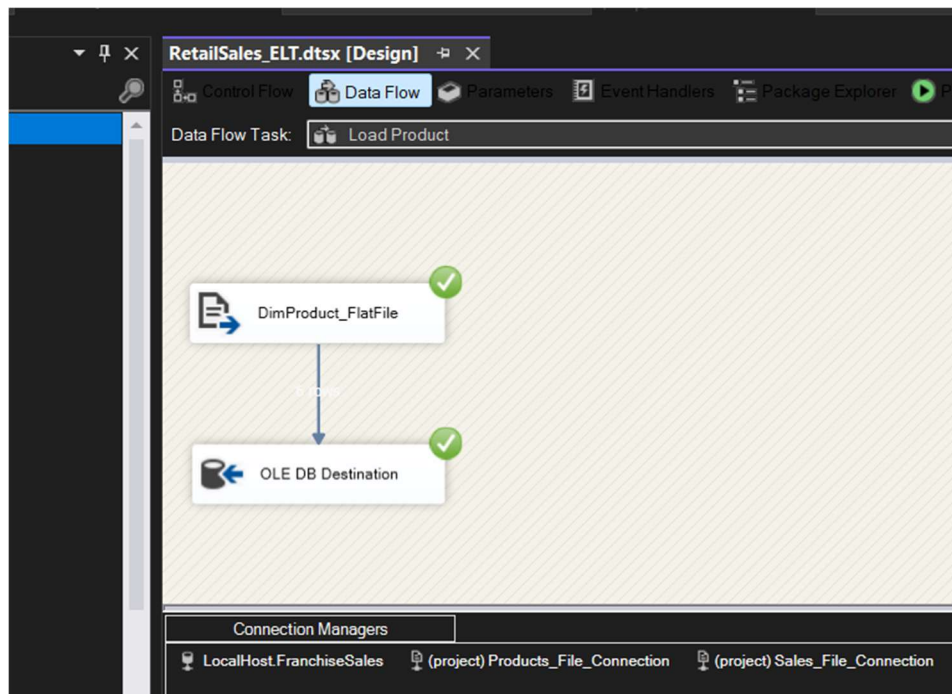
2.3 Product Destination Set (Mappings)

Mapped source columns to destination table fields to ensure accurate data loading into DimProduct.

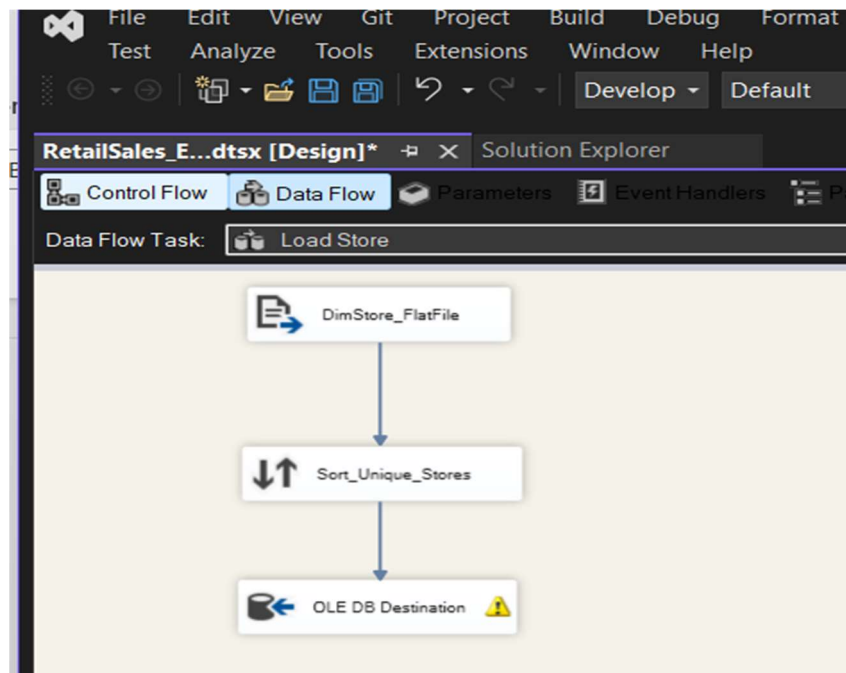


2.4 Load Product Run Successfully

Executed the data flow and confirmed that product data was loaded successfully without errors.

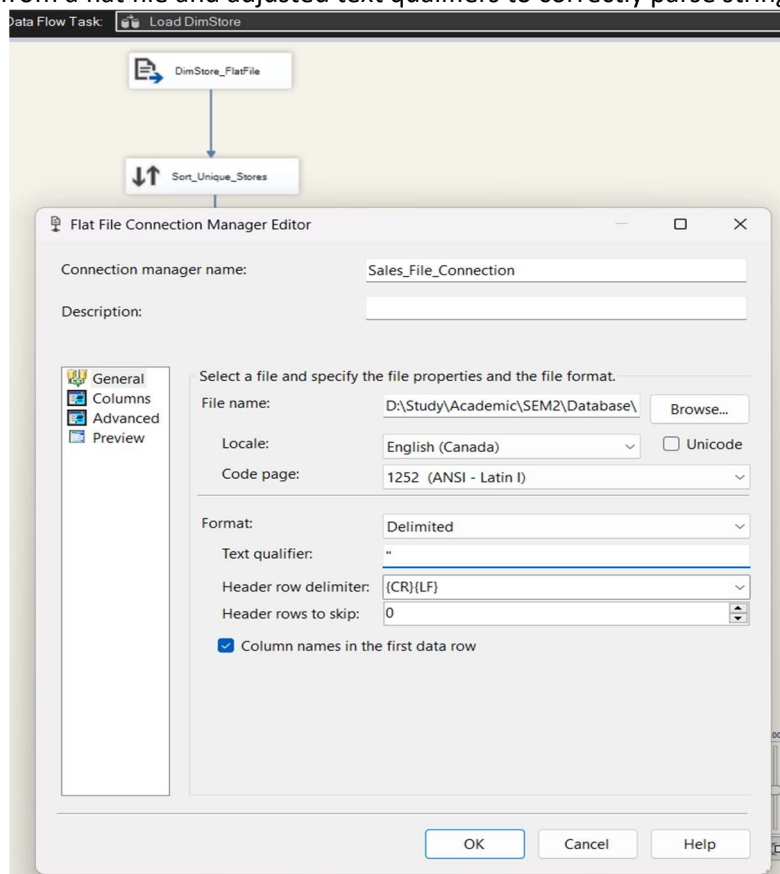


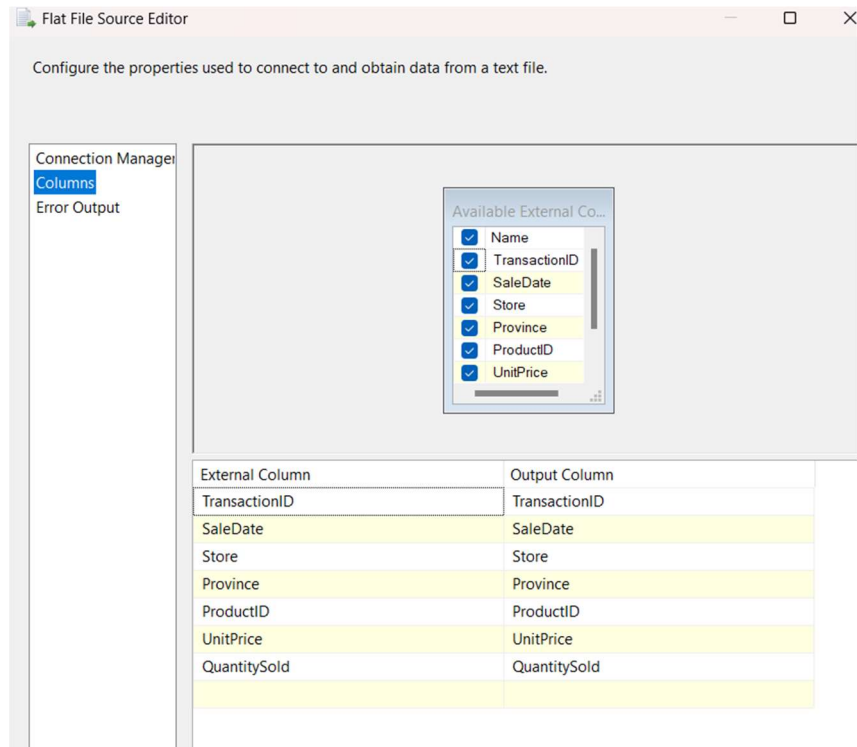
3.0 Control Flow-> Load Store



3.1 Store Flat File Load (Text Qualifier)

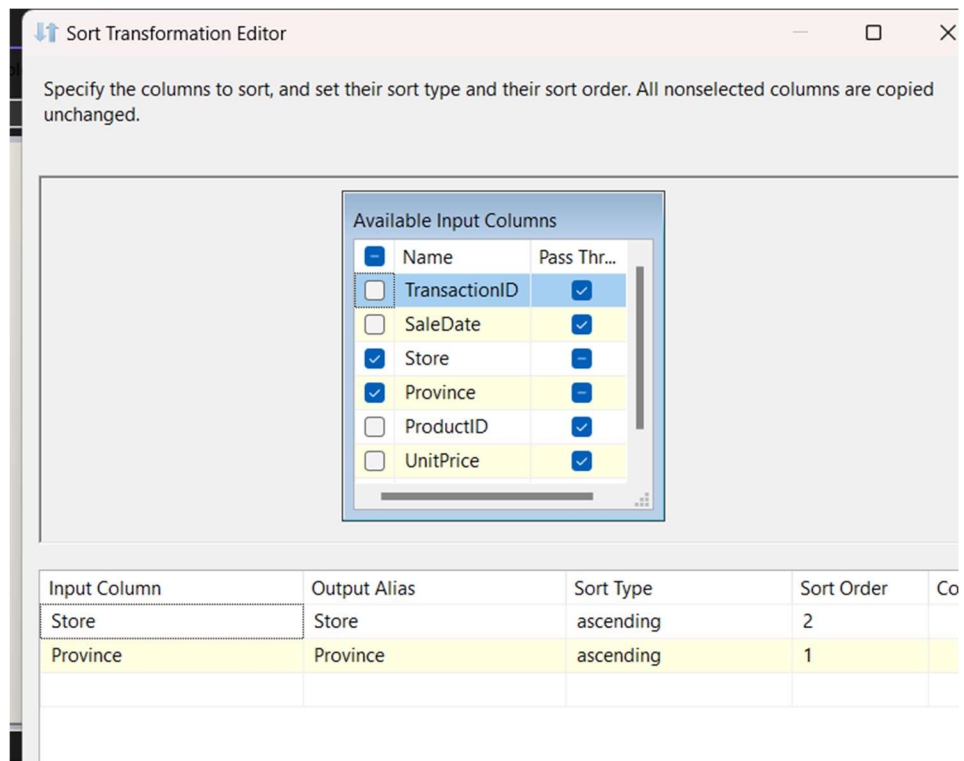
Loaded store data from a flat file and adjusted text qualifiers to correctly parse string values.





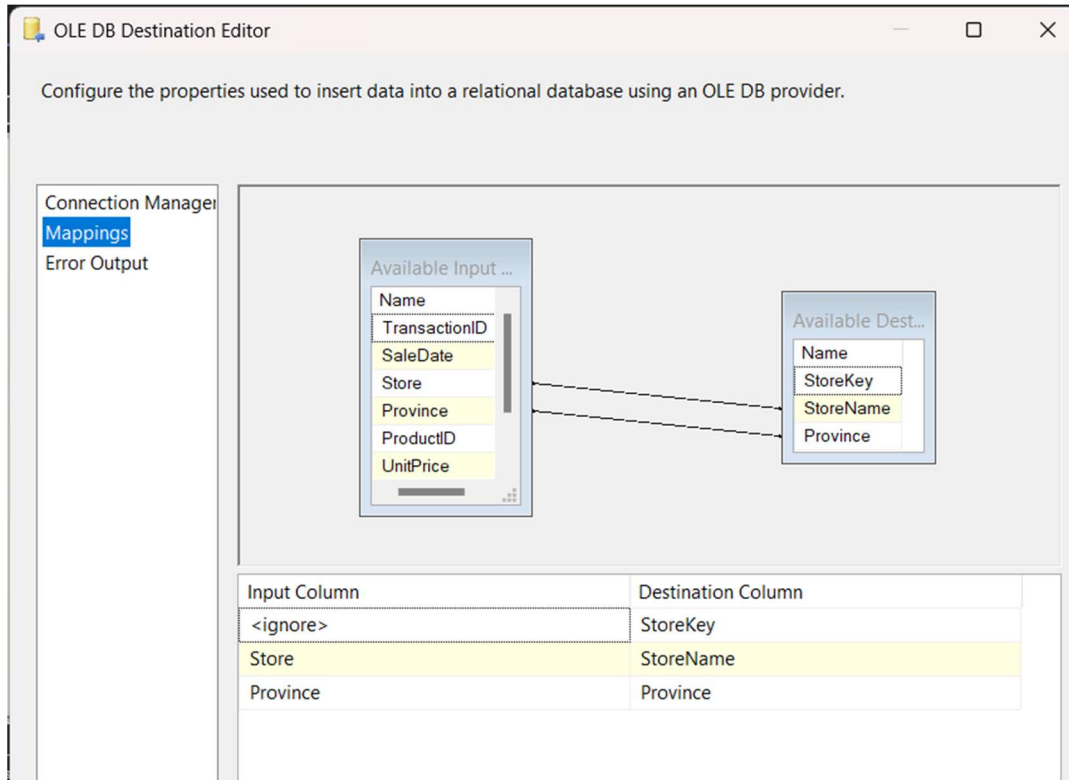
3.2 Sort Unique Stores

Applied sorting and removed duplicates to ensure only unique store records are loaded.



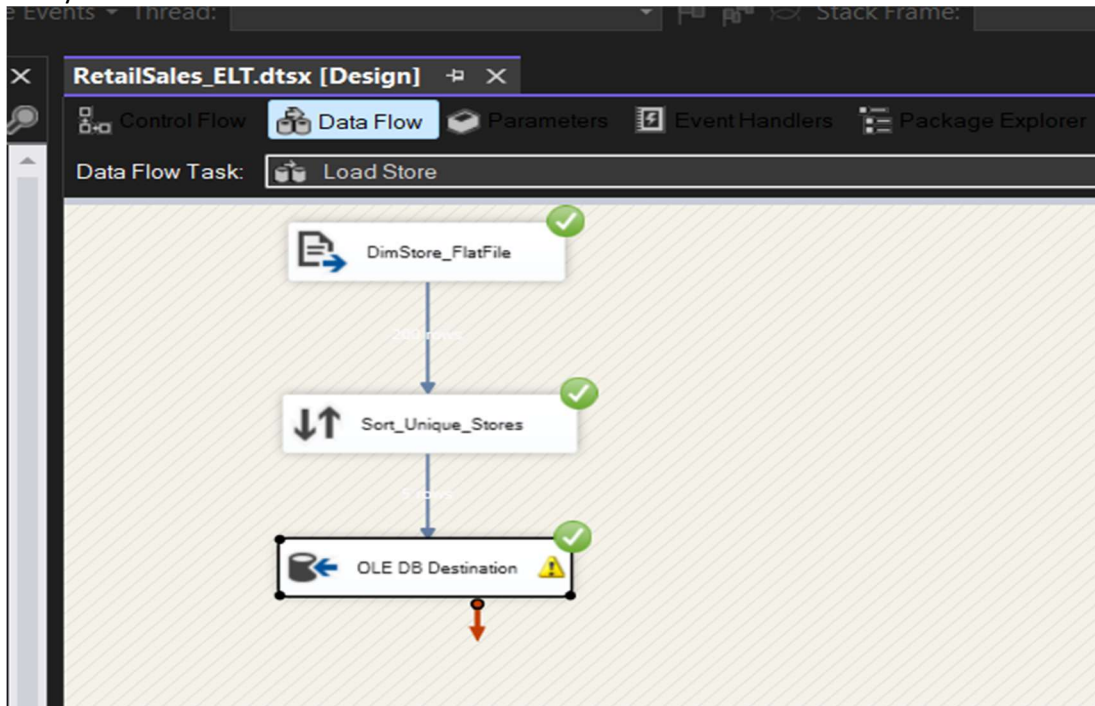
3.3 Store Destination (Mapping)

Mapped cleaned store data to the DimStore table structure.

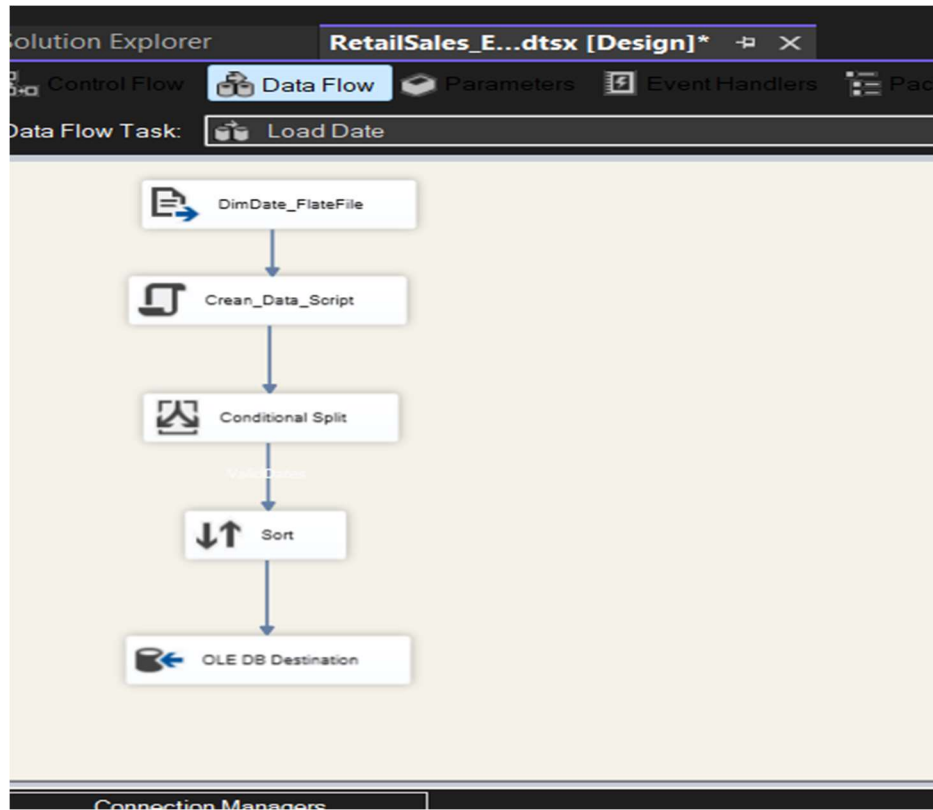


3.4 Load Store Run Successfully

Successfully executed the store data flow and validated correct data insertion.

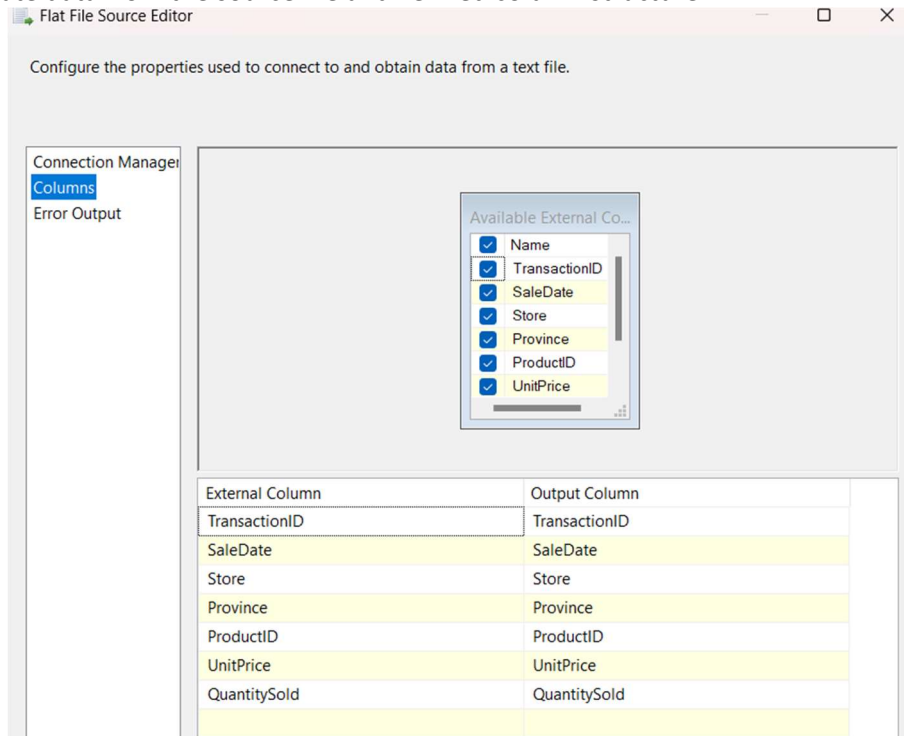


4.0 Control Flow-> Load Date



4.1 Flat File Column Load

Loaded raw date data from the source file and verified column structure.



4.2 Script for Date Cleaning

Implemented a script to clean and standardize date values.

```
public override void Input0_ProcessInputRow(Input0Buffer Row)
{
    DateTime parsedDate;

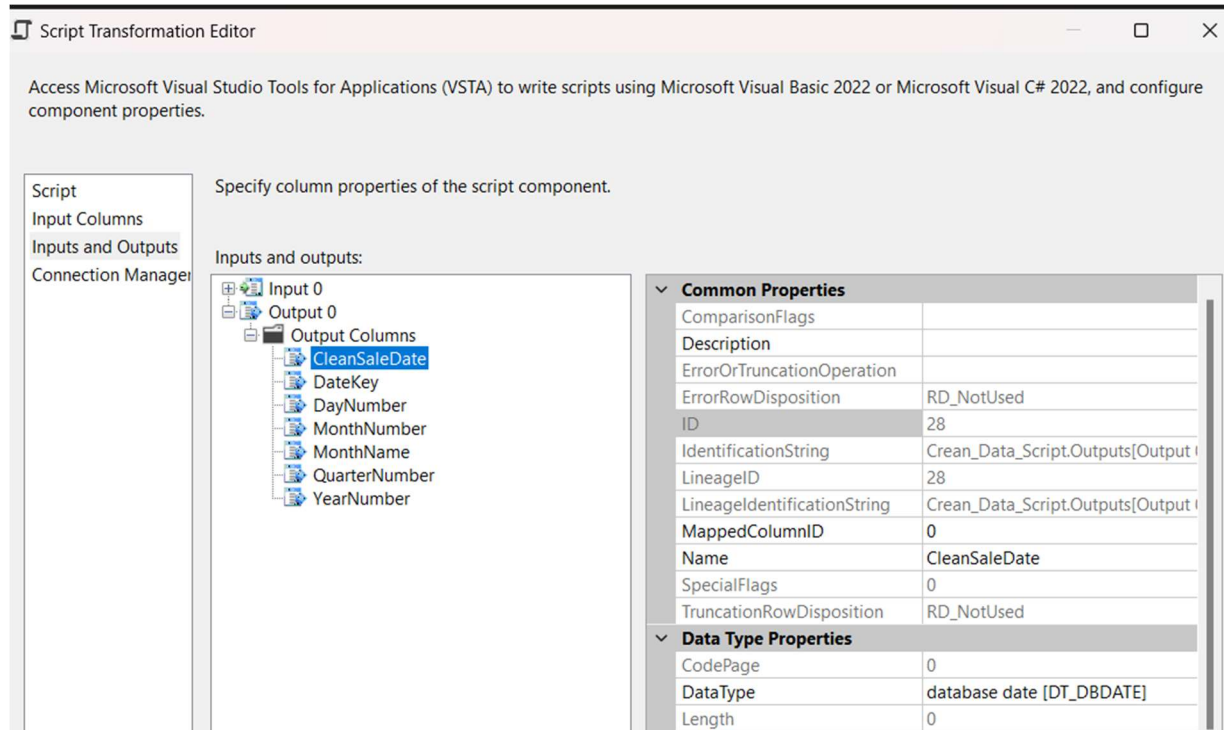
    if (!Row.SaleDate_IsNull &&
        DateTime.TryParse(Row.SaleDate.Trim(), out parsedDate))
    {
        Row.CleanSaleDate = parsedDate;
        Row.DateKey = parsedDate.Year * 10000 + parsedDate.Month * 100 + parsedDate.Day;
        Row.DayNumber = parsedDate.Day;
        Row.MonthNumber = parsedDate.Month;
        Row.MonthName = parsedDate.ToString("MMMM");
        Row.QuarterNumber = ((parsedDate.Month - 1) / 3) + 1;
        Row.YearNumber = parsedDate.Year;
    }
    else
    {
        Row.CleanSaleDate_IsNull = true;
        Row.DateKey_IsNull = true;
        Row.DayNumber_IsNull = true;
        Row.MonthNumber_IsNull = true;
        Row.MonthName_IsNull = true;
        Row.QuarterNumber_IsNull = true;
        Row.YearNumber_IsNull = true;
    }
}
```

4.3 Script Input and Output Setup

Configured input and output columns for the script component to process date data.

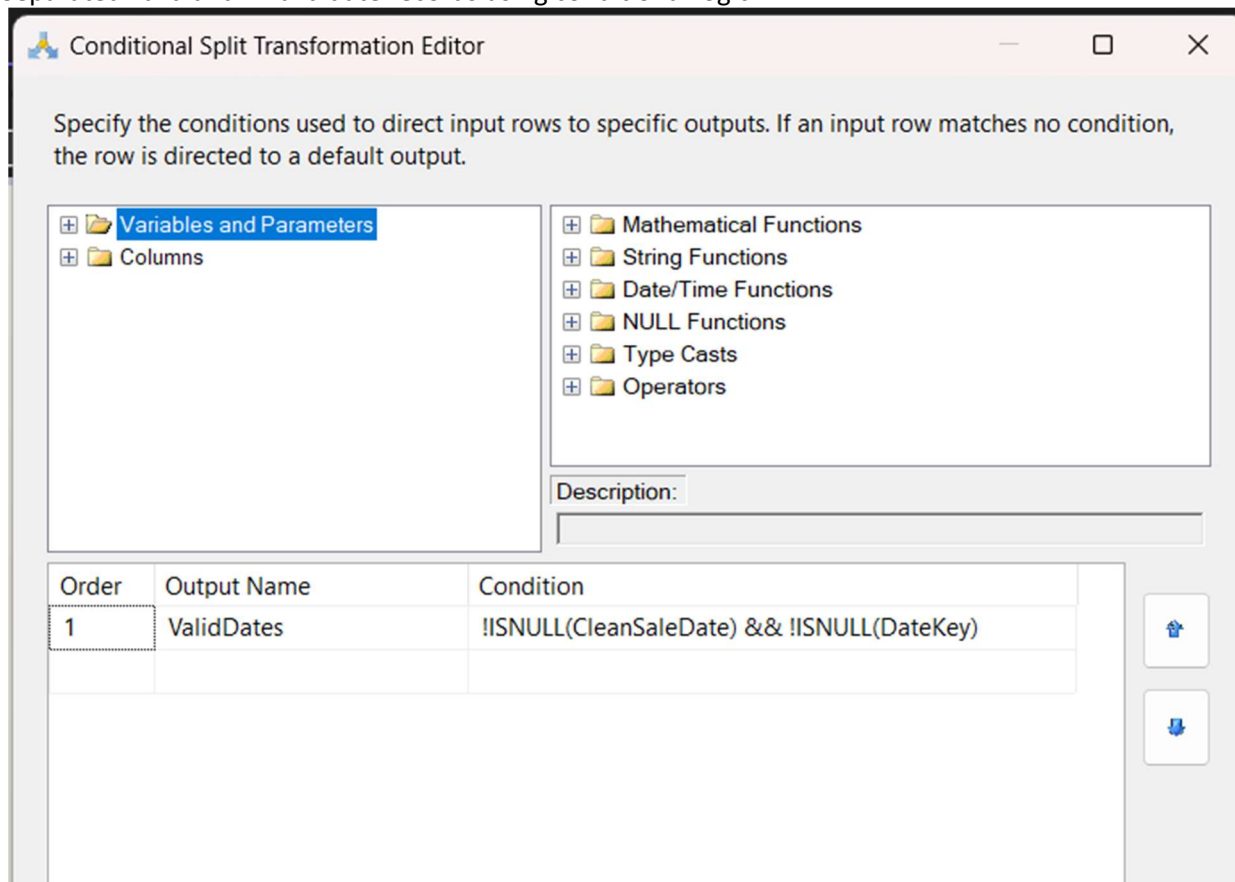
Access Microsoft Visual Studio Tools for Applications (VSTA) to write scripts using Microsoft Visual Basic 2022 or Microsoft Visual C# 2022, and configure component properties.

Input Column	Output Alias	Usage Type
SaleDate	SaleDate	ReadOnly



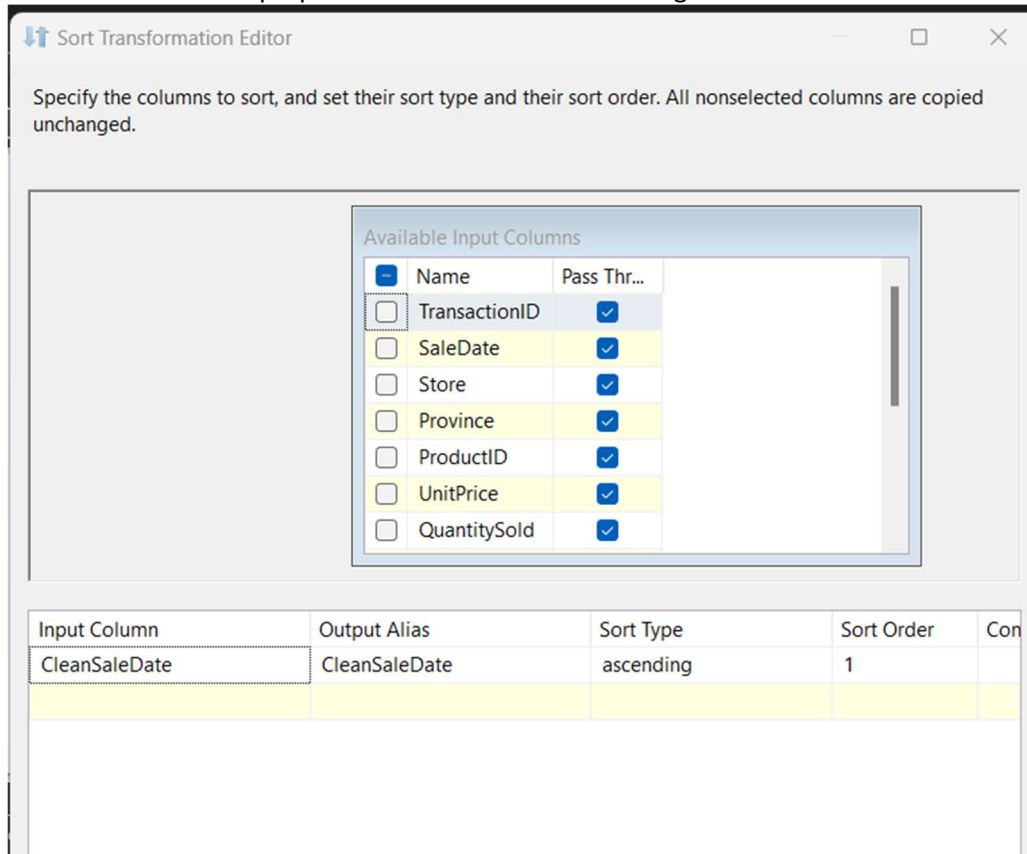
4.4 Conditional Split (Valid Dates)

Separated valid and invalid date records using conditional logic.



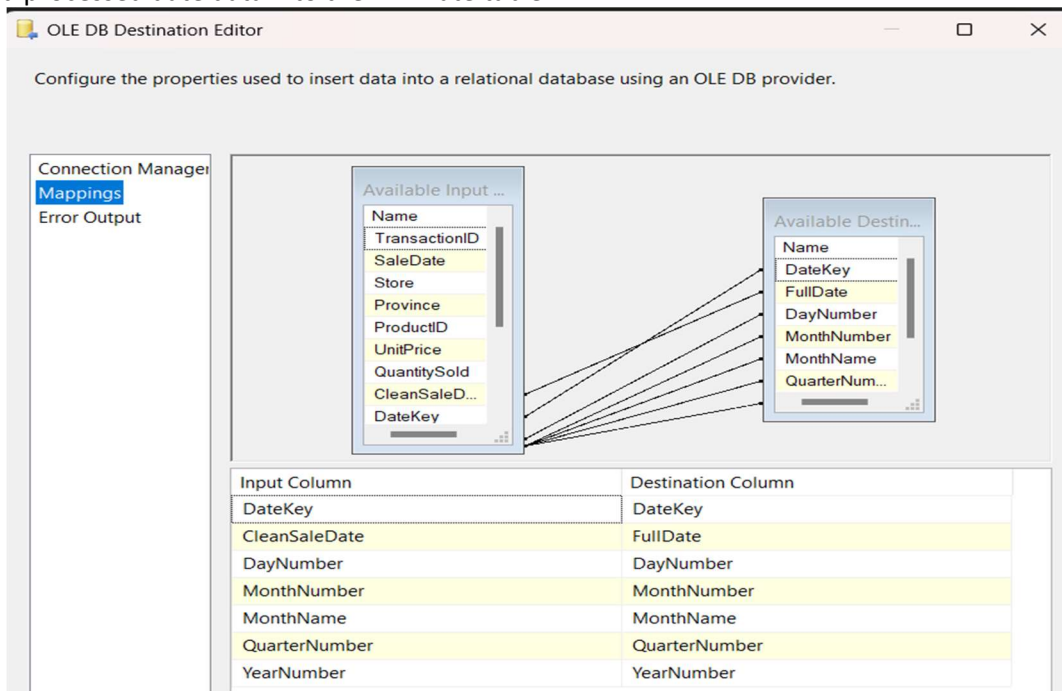
4.5 Sort (CleanSaleDate)

Sorted cleaned date values to prepare them for consistent loading.



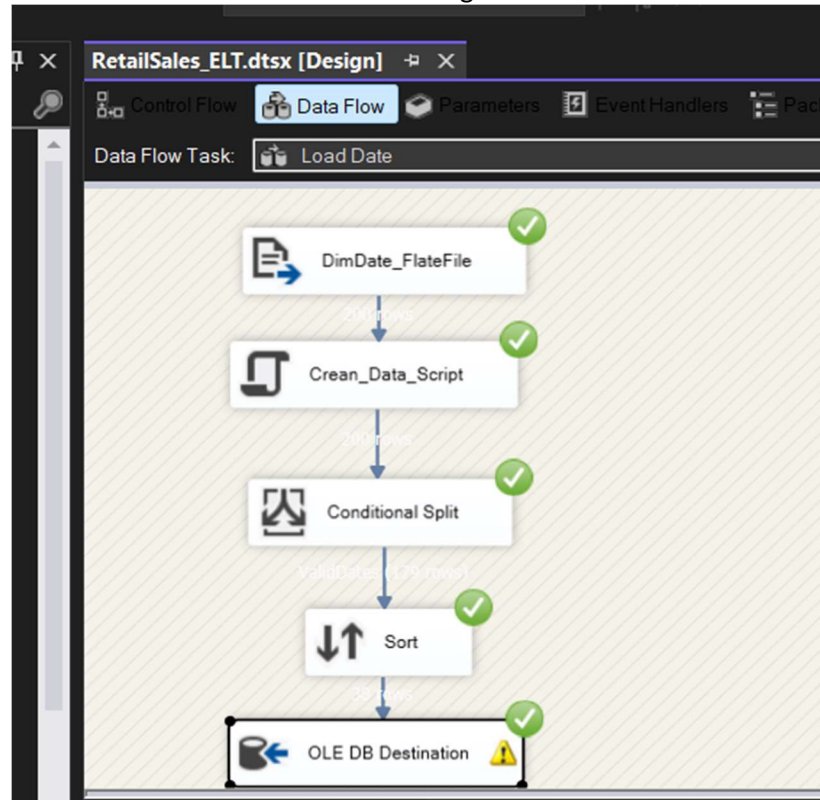
4.6 Destination for Date

Mapped processed date data into the DimDate table.

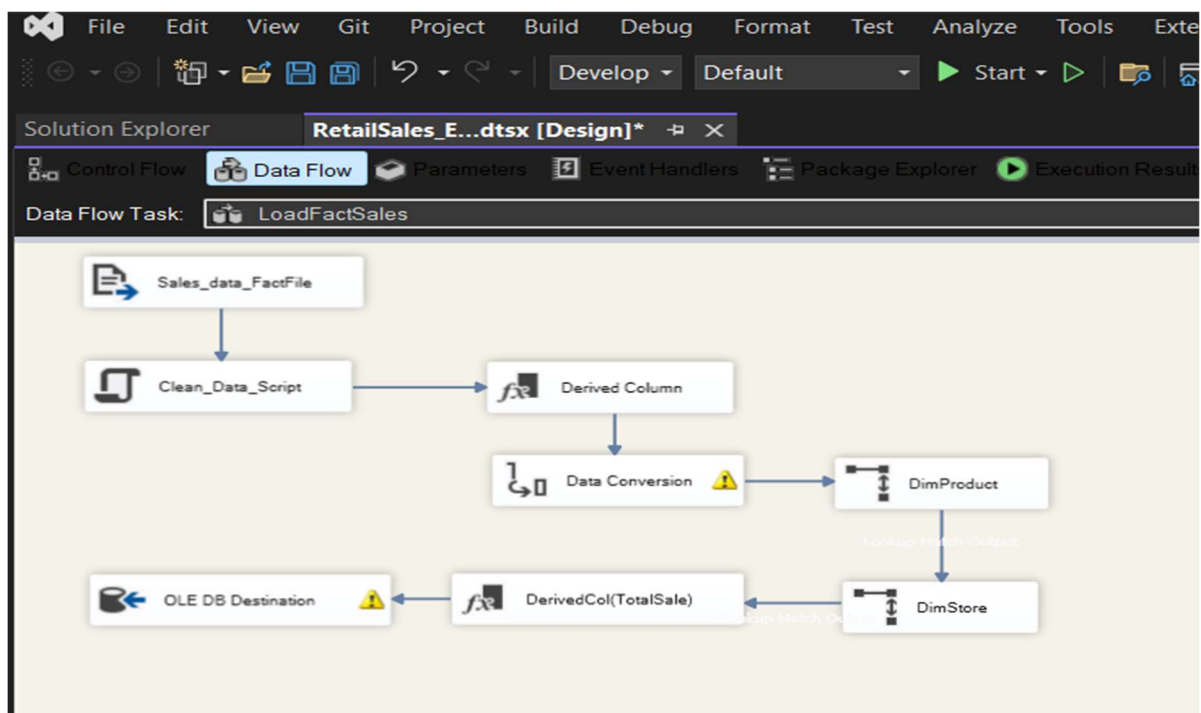


4.7 Date Data Flow Run Successfully

Executed the data flow and confirmed successful loading of date records.

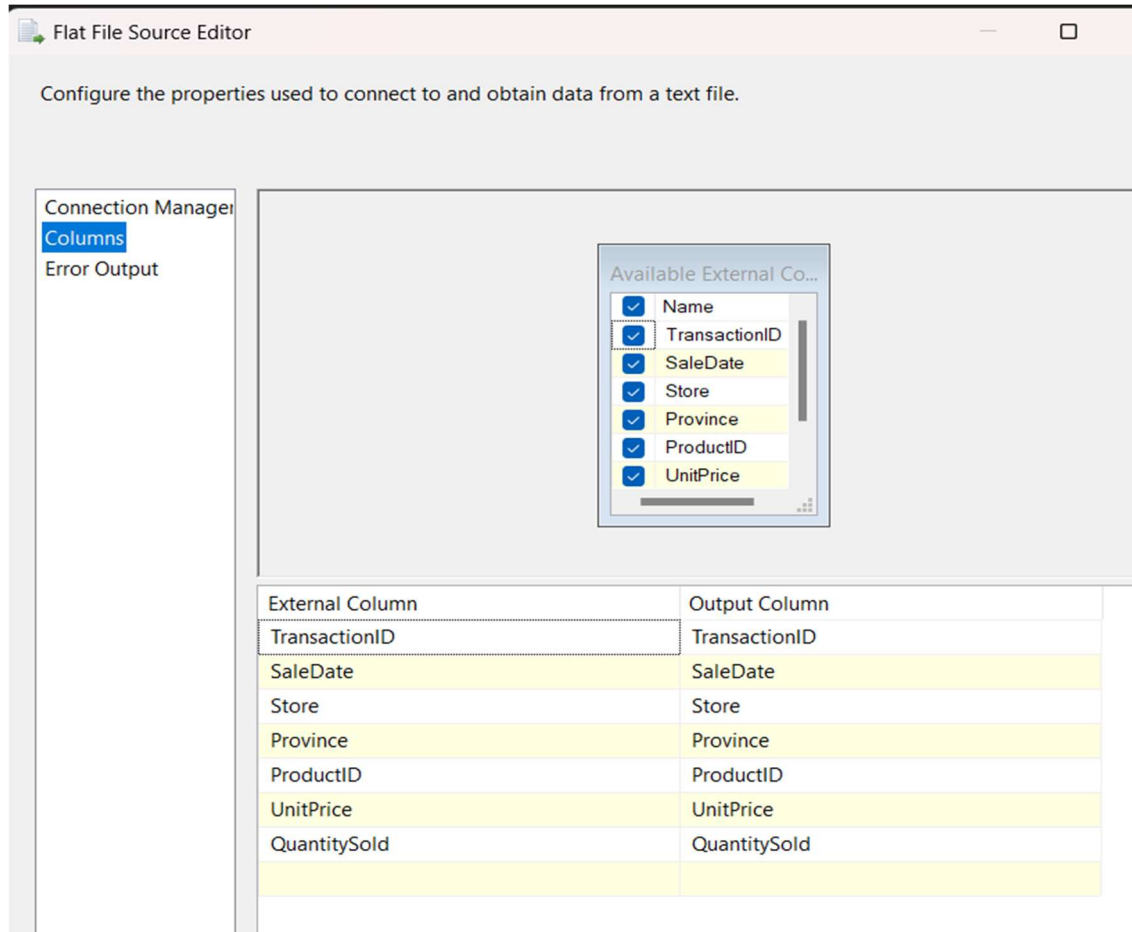


5.0 FactSales Data Flow



5.1 Flat File Column Load

Loaded raw sales data from the source file and verified column structure.



5.2 Clean Data Script (CleanSaleDate)

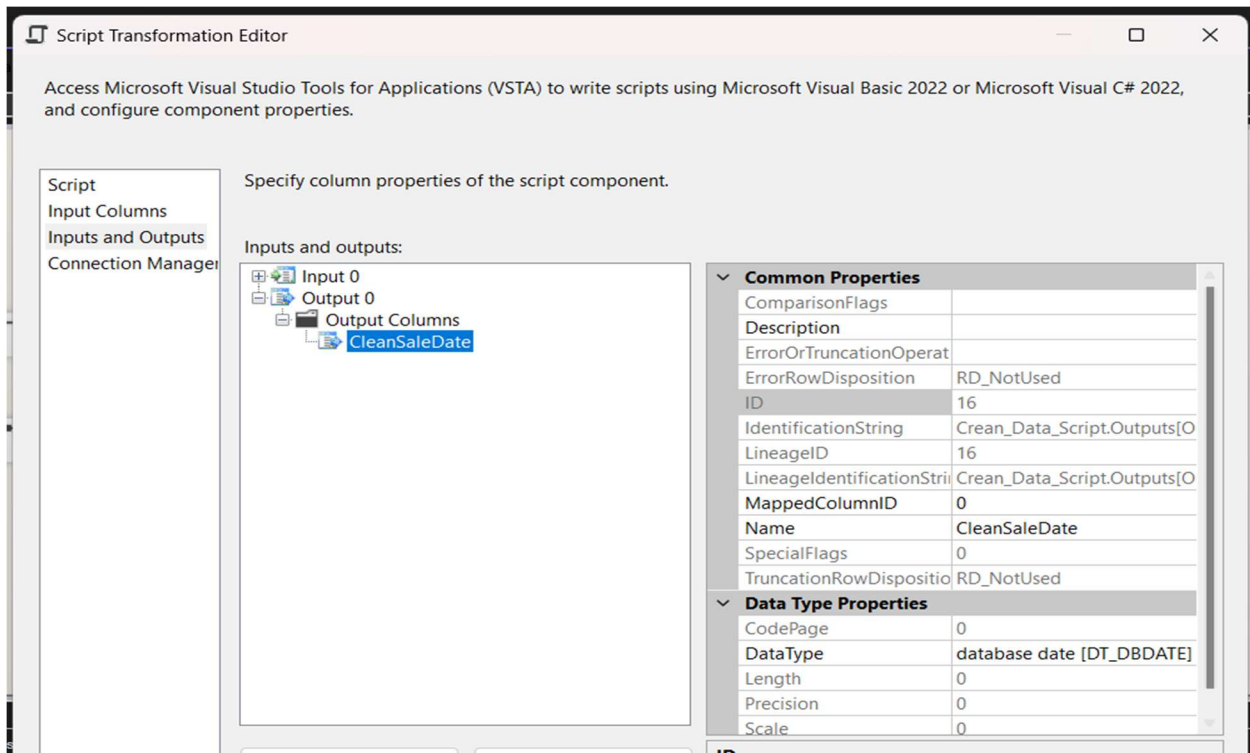
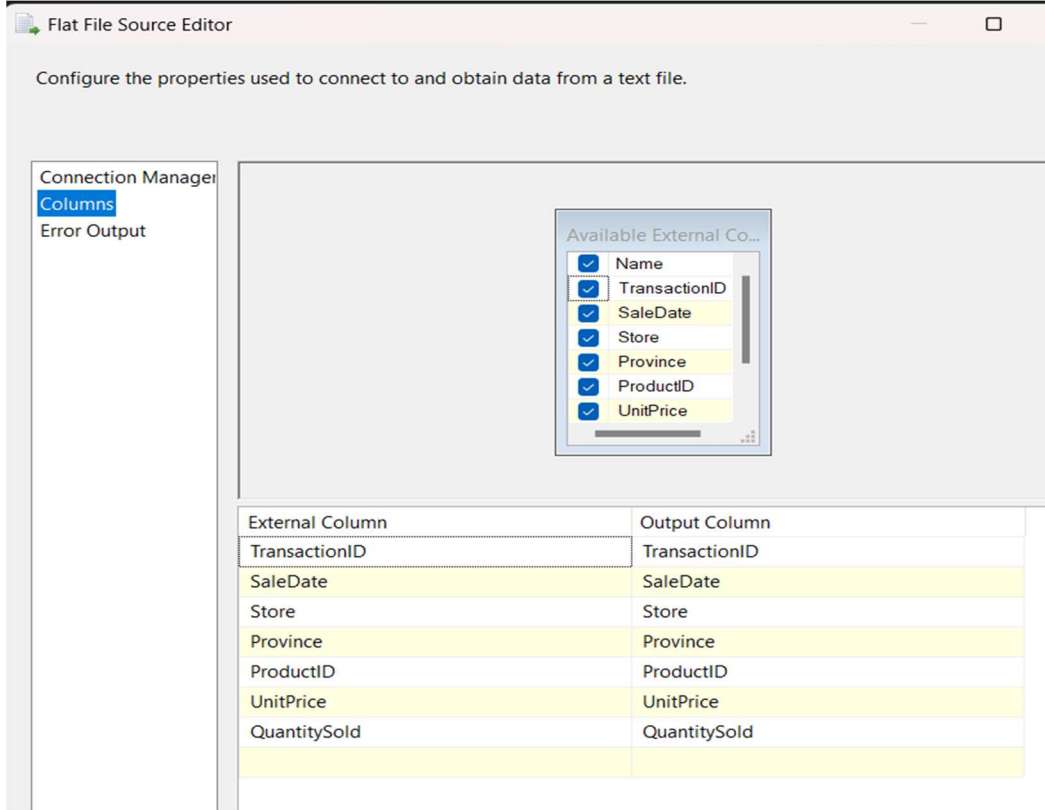
Applied a script to clean and standardize sales date values.

```
public override void Input0_ProcessInputRow(Input0Buffer Row)
{
    DateTime parsedDate;

    if (!Row.SaleDate_IsNull && DateTime.TryParse(Row.SaleDate.Trim(), out parsedDate))
    {
        Row.CleanSaleDate = parsedDate;
    }
    else
    {
        Row.CleanSaleDate_IsNull = true;
    }
}
```

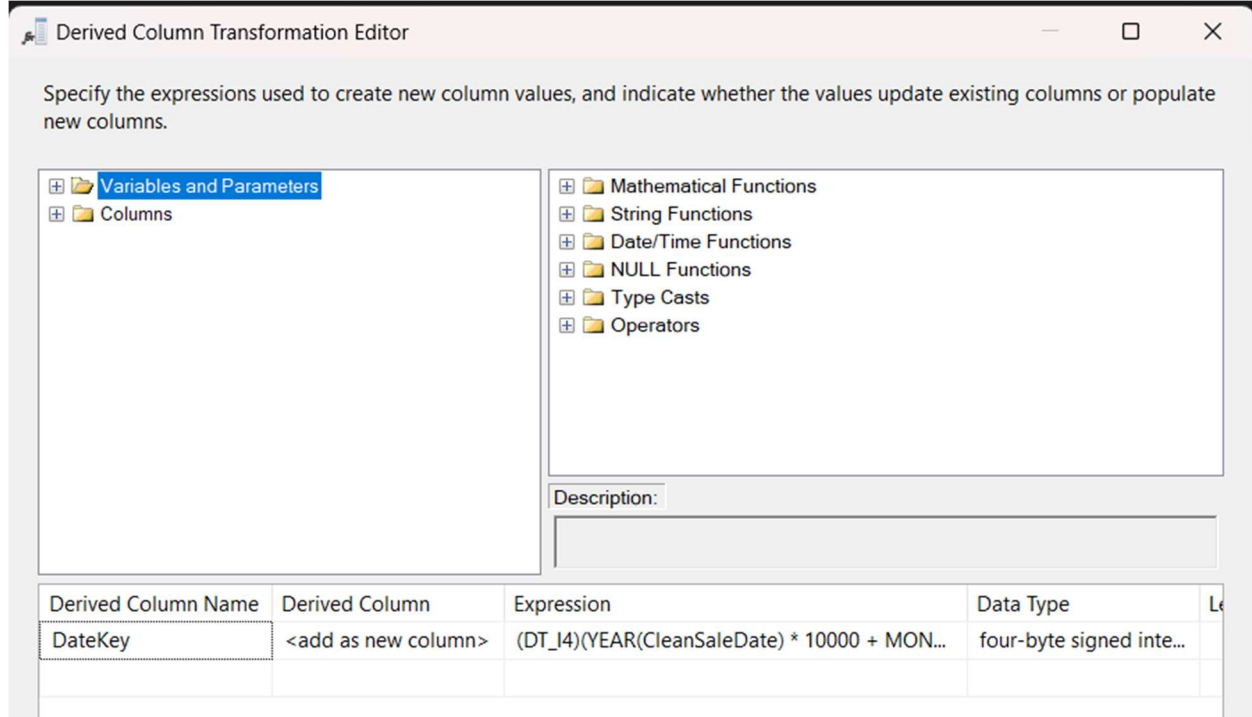
5.3 Script Input and Output Setup

Configured script component inputs and outputs for processing sales data.



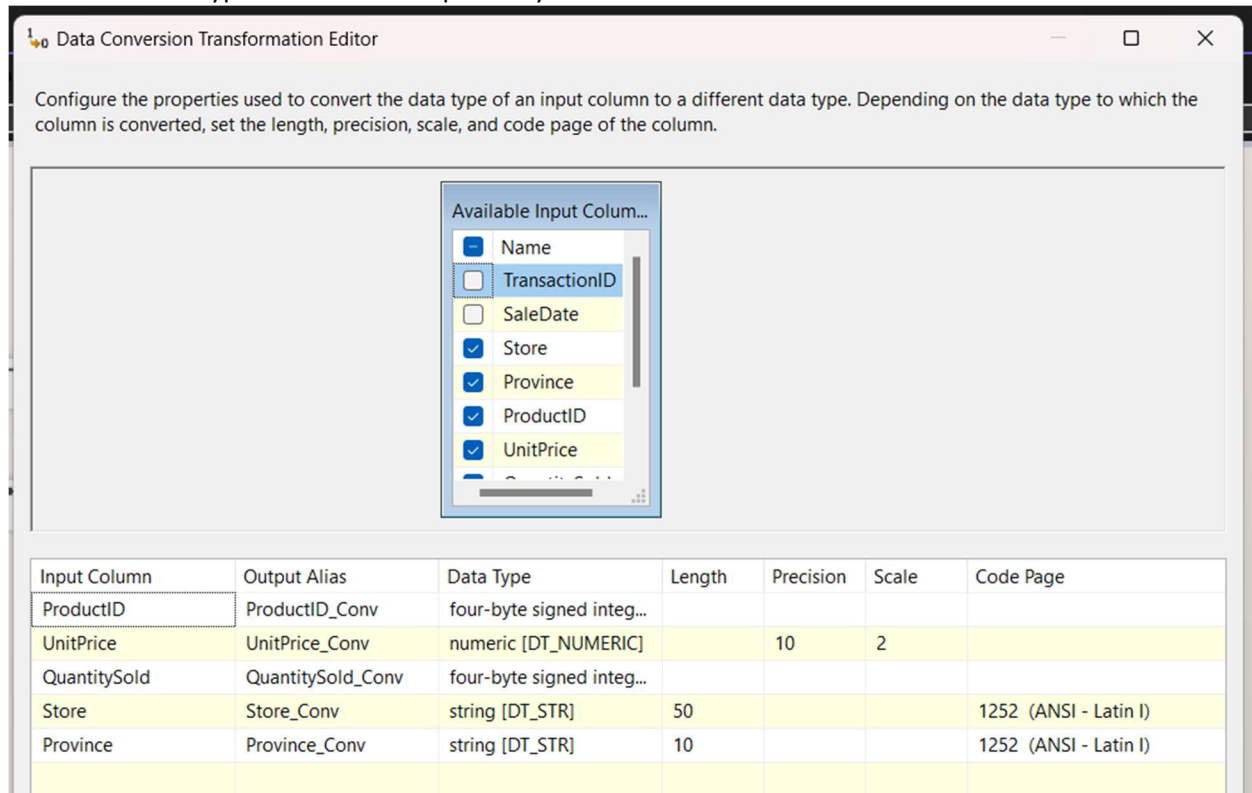
5.4 Derived Column (DateKey)

Created a DateKey column to match the DimDate table format.



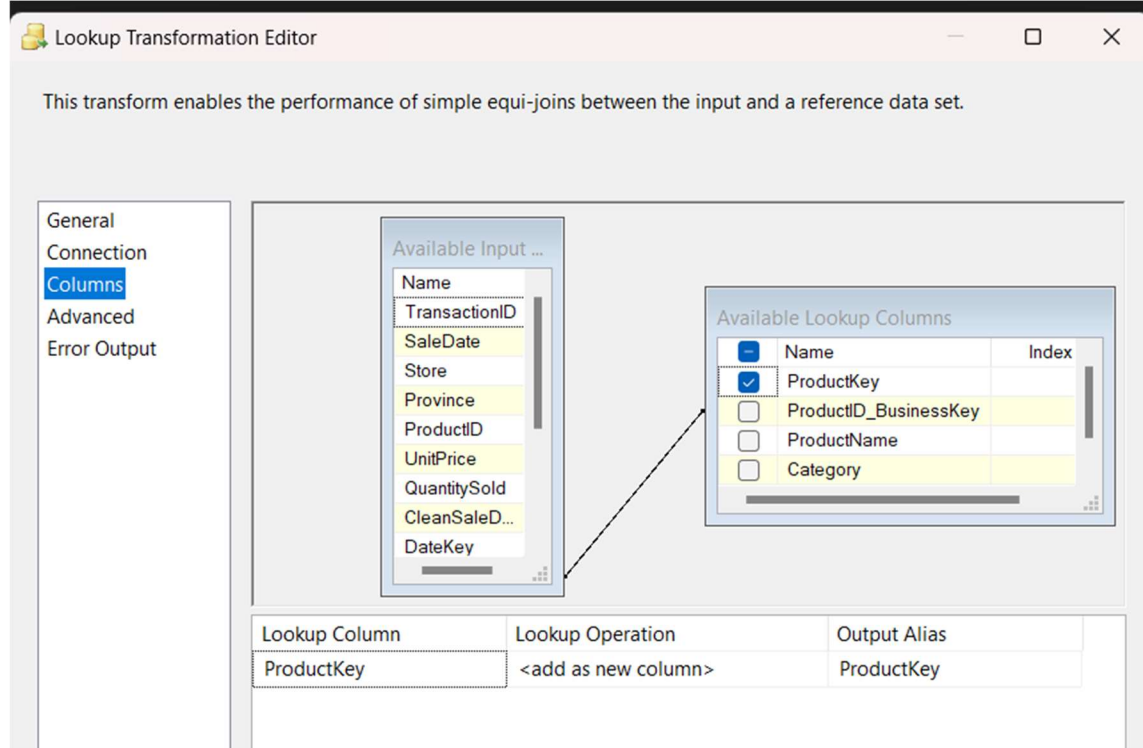
5.5 Data Conversion

Converted data types to ensure compatibility with destination table schema.



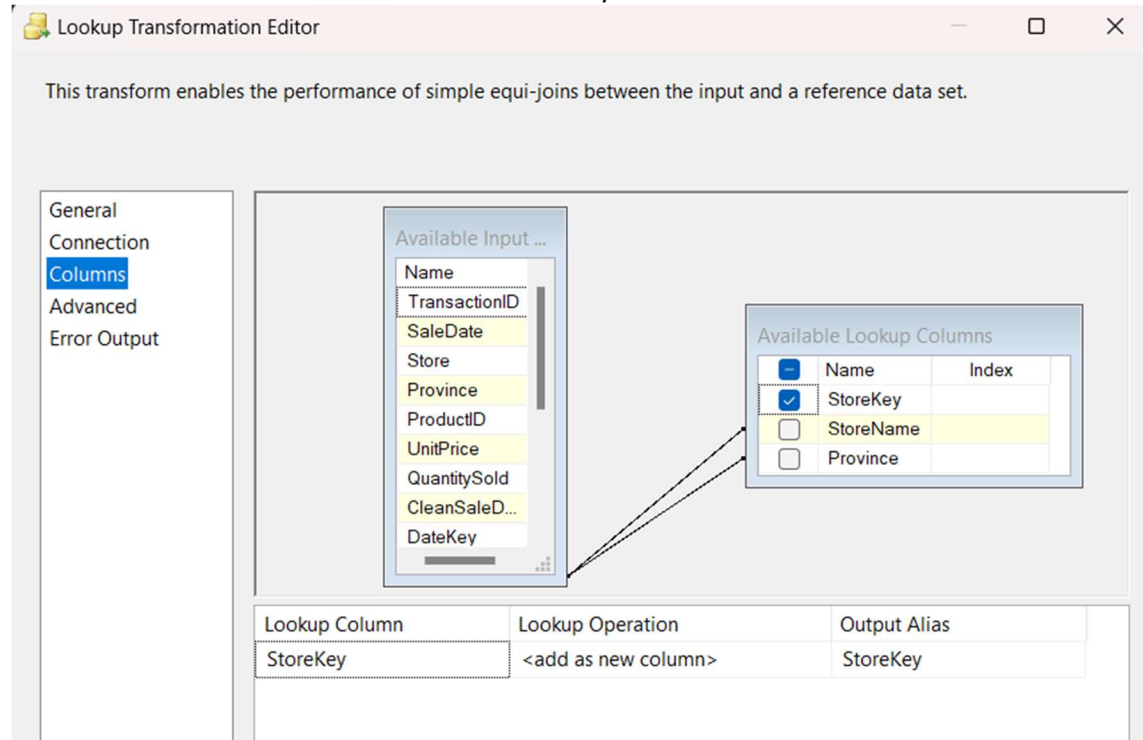
5.6 Lookup for Product Data

Matched sales data with DimProduct to retrieve ProductKey.



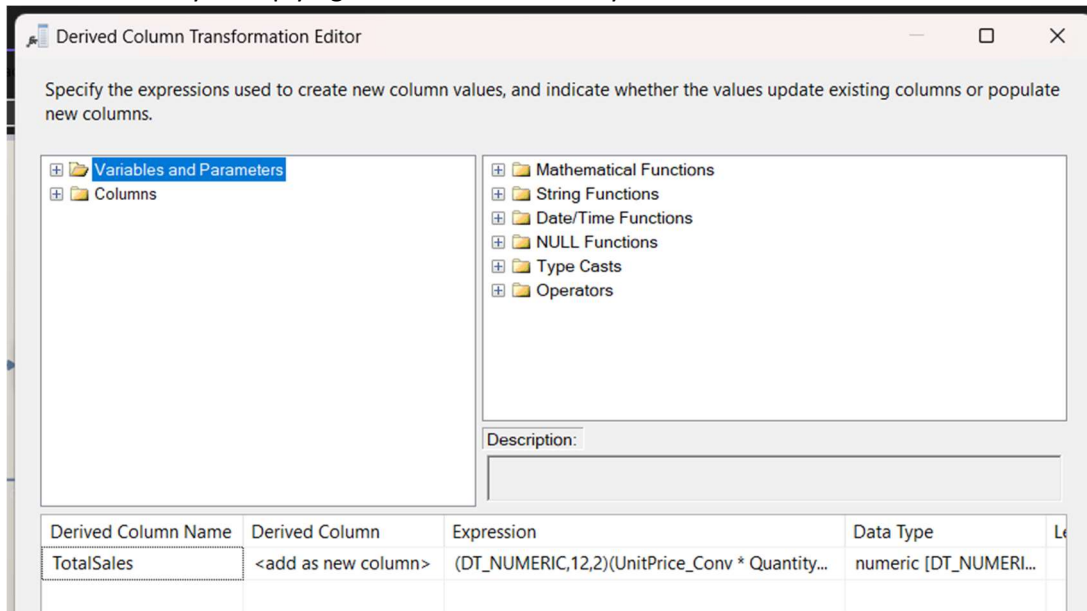
5.7 Lookup for Store Data

Matched sales data with DimStore to retrieve StoreKey.



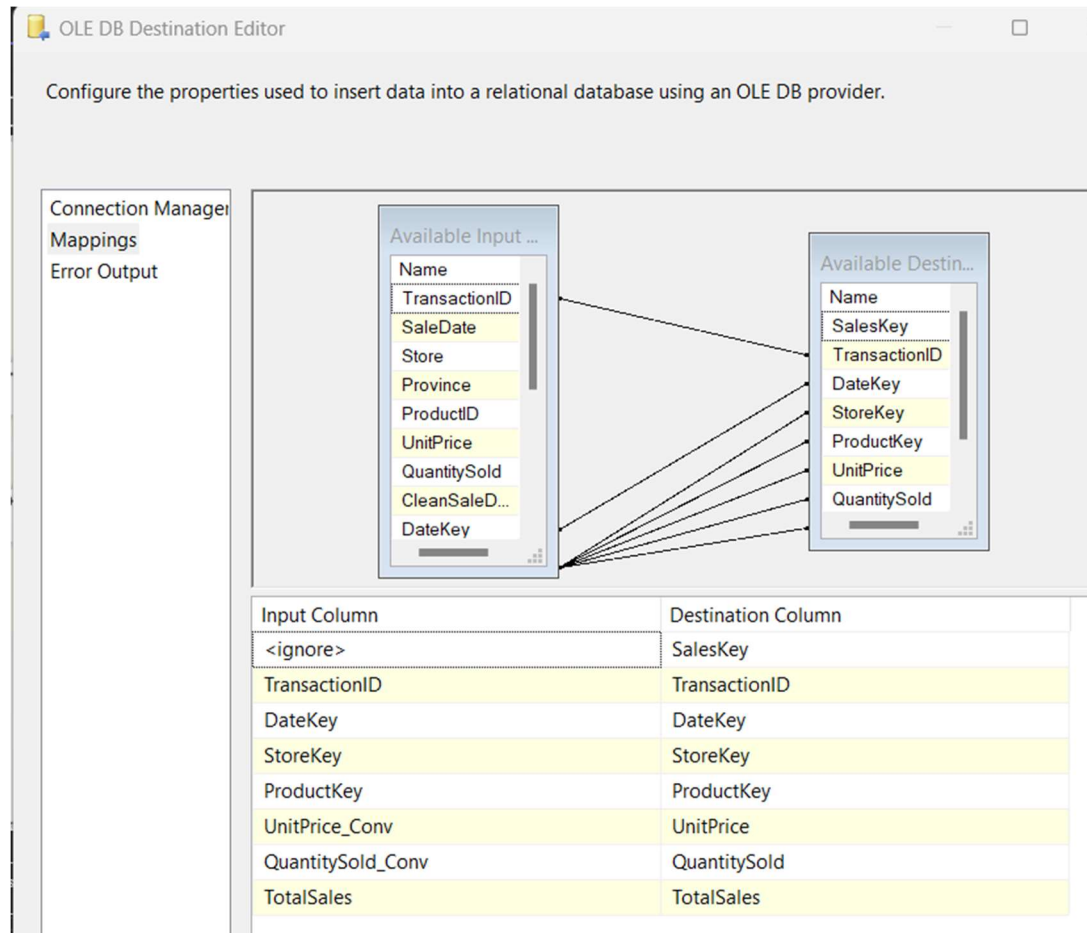
5.8 Derived Column (TotalSale)

Calculated TotalSale by multiplying UnitPrice and QuantitySold.



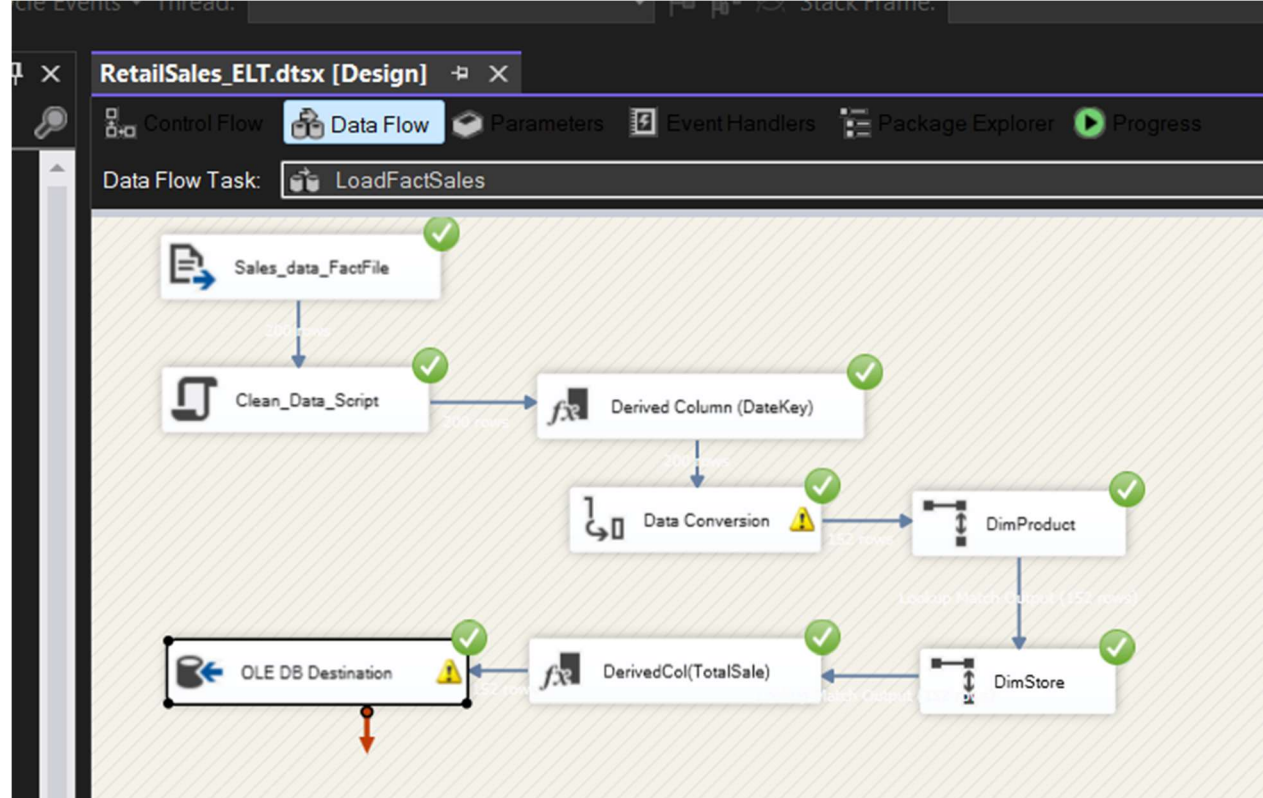
5.9 FactSales Destination (Mapping)

Mapped transformed data to the FactSales table.



5.10 FactSales Run Successfully

Executed the data flow and confirmed successful loading of sales data.



5.11 DimProduct Table in SQL

SQLQuery1.s...\User (78))* ✕

1
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SELECT * FROM DimProduct
SELECT * FROM DimStore
SELECT * FROM DimDate
SELECT * FROM FactSales

108 %
No issues found

Results Messages

	SalesKey	TransactionID	DateKey	StoreKey	ProductKey	UnitPrice	QuantitySold	TotalSales
1	1	1	20250117	24	32	129.76	8	1038.08
2	2	2	20250125	21	31	1491.65	1	1491.65
3	3	4	20250104	24	35	85.73	2	171.46
4	4	5	20250111	24	32	129.76	8	1038.08
5	5	6	20250116	23	35	85.73	2	171.46
6	6	7	20250108	21	31	1491.65	2	2983.30
7	7	9	20250111	23	31	1491.65	1	1491.65
8	8	10	20250109	23	35	85.73	10	857.30
9	9	11	20250127	24	31	1491.65	8	11933.20
10	10	13	20250106	21	36	806.10	8	6448.80
11	11	16	20250106	23	36	806.10	2	1612.20
12	12	18	20250119	22	36	806.10	3	2418.30
13	13	19	20250128	22	33	143.93	3	431.79
14	14	20	20250118	21	32	129.76	10	1297.60
15	15	22	20250110	22	35	85.73	10	857.30
16	16	30	20250101	25	36	806.10	1	806.10
17	17	31	20250106	24	35	85.73	3	257.19
18	18	32	20250126	21	36	806.10	7	5642.70
19	19	33	20250123	24	33	143.93	4	575.72
20	20	35	20250128	24	36	806.10	4	3224.40
21	21	36	20250107	24	31	1491.65	4	5966.60
22	22	39	20250126	23	35	85.73	3	257.19
23	23	41	20250111	23	36	806.10	1	806.10
24	24	42	20250124	21	31	1491.65	4	5966.60
25	25	43	20250128	21	35	85.73	4	342.92
26	26	44	20250110	21	35	85.73	3	257.19
27	27	46	20250102	23	36	806.10	6	4836.60
28	28	47	20250129	21	35	85.73	9	771.57
29	29	48	20250104	23	33	143.93	1	143.93
30	30	49	20250109	21	32	129.76	6	778.56
31	31	50	20250113	23	35	85.73	9	771.57
32	32	51	20250125	21	31	1491.65	5	7458.25
33	33	52	20250131	22	32	129.76	2	259.52
34	34	55	20250125	24	35	85.73	4	342.92
35	35	57	20250107	21	34	30.43	6	182.58
36	36	58	20250111	21	35	85.73	6	514.38
37	37	59	20250116	24	35	85.73	3	257.19
38	38	60	20250123	21	31	1491.65	9	13424.85
39	39	61	20250110	24	35	85.73	1	85.73
40	40	62	20250106	21	35	85.73	3	257.19
41	41	63	20250116	21	32	129.76	7	908.32
42	42	64	20250111	24	32	129.76	10	1297.60

5.12 DimStore Table in SQL

Execute

SQLQuery1.s... \User (78))*

```
1 SELECT * FROM DimProduct
2 SELECT * FROM DimStore
3 SELECT * FROM DimDate
4 SELECT * FROM FactSales
```

108 % No issues found

Results Messages

	StoreKey	StoreName	Province
1	21	Calgary	AB
2	22	Edmonton	AB
3	23	Winnipeg	MB
4	24	Regina	SK
5	25	Saskatoon	SK
6	26	Calgary	AB
7	27	Edmonton	AB
8	28	Winnipeg	MB
9	29	Regina	SK
10	30	Saskatoon	SK
11	31	Calgary	AB
12	32	Edmonton	AB
13	33	Winnipeg	MB
14	34	Regina	SK
15	35	Saskatoon	SK
16	36	Calgary	AB
17	37	Edmonton	AB
18	38	Winnipeg	MB
19	39	Regina	SK
20	40	Saskatoon	SK
21	41	Calgary	AB
22	42	Edmonton	AB
23	43	Winnipeg	MB
24	44	Regina	SK
25	45	Saskatoon	SK
26	46	Calgary	AB
27	47	Edmonton	AB
28	48	Winnipeg	MB
29	49	Regina	SK
30	50	Saskatoon	SK
31	51	Calgary	AB
32	52	Edmonton	AB
33	53	Winnipeg	MB
34	54	Regina	SK
35	55	Saskatoon	SK

5.13 DimDate Table in SQL

Execute

SQLQuery1.s... \User (78))*

```
1 SELECT * FROM DimProduct
2 SELECT * FROM DimStore
3 SELECT * FROM DimDate
4 SELECT * FROM FactSales
```

108 % No issues found

Results Messages

	DateKey	FullDate	DayNumber	MonthNumber	MonthName	QuarterNumber	YearNumber
1	20250101	2025-01-01	1	1	January	1	2025
2	20250102	2025-01-02	2	1	January	1	2025
3	20250103	2025-01-03	3	1	January	1	2025
4	20250104	2025-01-04	4	1	January	1	2025
5	20250105	2025-01-05	5	1	January	1	2025
6	20250106	2025-01-06	6	1	January	1	2025
7	20250107	2025-01-07	7	1	January	1	2025
8	20250108	2025-01-08	8	1	January	1	2025
9	20250109	2025-01-09	9	1	January	1	2025
10	20250110	2025-01-10	10	1	January	1	2025
11	20250111	2025-01-11	11	1	January	1	2025
12	20250112	2025-01-12	12	1	January	1	2025
13	20250113	2025-01-13	13	1	January	1	2025
14	20250114	2025-01-14	14	1	January	1	2025
15	20250115	2025-01-15	15	1	January	1	2025
16	20250116	2025-01-16	16	1	January	1	2025
17	20250117	2025-01-17	17	1	January	1	2025
18	20250118	2025-01-18	18	1	January	1	2025
19	20250119	2025-01-19	19	1	January	1	2025
20	20250120	2025-01-20	20	1	January	1	2025
21	20250121	2025-01-21	21	1	January	1	2025
22	20250122	2025-01-22	22	1	January	1	2025
23	20250123	2025-01-23	23	1	January	1	2025
24	20250124	2025-01-24	24	1	January	1	2025
25	20250125	2025-01-25	25	1	January	1	2025
26	20250126	2025-01-26	26	1	January	1	2025
27	20250127	2025-01-27	27	1	January	1	2025
28	20250128	2025-01-28	28	1	January	1	2025
29	20250129	2025-01-29	29	1	January	1	2025
30	20250130	2025-01-30	30	1	January	1	2025
31	20250131	2025-01-31	31	1	January	1	2025
32	20250201	2025-02-01	1	2	February	1	2025
33	20250301	2025-03-01	1	3	March	1	2025
34	20250501	2025-05-01	1	5	May	2	2025
35	20250701	2025-07-01	1	7	July	3	2025
36	20250801	2025-08-01	1	8	August	3	2025
37	20251101	2025-11-01	1	11	November	4	2025
38	20251201	2025-12-01	1	12	December	4	2025

5.14 FactSales Table in SQL

Execute

SQLQuery1.s...\User (78))*

1

2

3

4

SELECT * FROM DimProduct

SELECT * FROM DimStore

SELECT * FROM DimDate

SELECT * FROM FactSales

108 %

No issues found

Results

Messages

	SalesKey	TransactionID	DateKey	StoreKey	ProductKey	UnitPrice	QuantitySold	TotalSales	
1	1	1	20250117	24	32	129.76	8	1038.08	
2	2	2	20250125	21	31	1491.65	1	1491.65	
3	3	4	20250104	24	35	85.73	2	171.46	
4	4	5	20250111	24	32	129.76	8	1038.08	
5	5	6	20250116	23	35	85.73	2	171.46	
6	6	7	20250108	21	31	1491.65	2	2983.30	
7	7	9	20250111	23	31	1491.65	1	1491.65	
8	8	10	20250109	23	35	85.73	10	857.30	
9	9	11	20250127	24	31	1491.65	8	11933.20	
10	10	13	20250106	21	36	806.10	8	6448.80	
11	11	16	20250106	23	36	806.10	2	1612.20	
12	12	18	20250119	22	36	806.10	3	2418.30	
13	13	19	20250128	22	33	143.93	3	431.79	
14	14	20	20250118	21	32	129.76	10	1297.60	
15	15	22	20250110	22	35	85.73	10	857.30	
16	16	30	20250101	25	36	806.10	1	806.10	
17	17	31	20250106	24	35	85.73	3	257.19	
18	18	32	20250126	21	36	806.10	7	5642.70	
19	19	33	20250123	24	33	143.93	4	575.72	
20	20	35	20250128	24	36	806.10	4	3224.40	
21	21	36	20250107	24	31	1491.65	4	5966.60	
22	22	39	20250126	23	35	85.73	3	257.19	
23	23	41	20250111	23	36	806.10	1	806.10	
24	24	42	20250124	21	31	1491.65	4	5966.60	
25	25	43	20250128	21	35	85.73	4	342.92	
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29	29	48	20250104	23	33	143.93	1	143.93	
30	30	49	20250109	21	32	129.76	6	778.56	
31	31	50	20250113	23	35	85.73	9	771.57	
32	32	51	20250125	21	31	1491.65	5	7458.25	
33	33	52	20250131	22	32	129.76	2	259.52	
34	34	55	20250125	24	35	85.73	4	342.92	
35	35	57	20250107	21	34	30.43	6	182.58	
36	36	58	20250111	21	35	85.73	6	514.38	
37	37	59	20250116	24	35	85.73	3	257.19	
38	38	60	20250123	21	31	1491.65	9	13424.85	
39	39	61	20250110	24	35	85.73	1	85.73	
40	40	62	20250106	21	35	85.73	3	257.19	
41	41	63	20250116	21	32	129.76	7	908.32	
42	42	64	20250111	24	33	143.93	10	1439.30	